

QUESTION ONE (15 Marks)

Marks

- (a) Evaluate $\frac{\sqrt{24.1 - 15.8}}{14 \times 23 - 25}$, correct to 4 significant figures. 2
- (b) Simplify:
- (i) $(2b^3)^4$ 1
- (ii) $y^{\frac{1}{2}} \times y^{\frac{-1}{2}}$ 2
- (c) Write without fractional or negative indices:
- (i) $x^{\frac{2}{3}}$ 1
- (ii) $(5x - 3)^{\frac{-1}{2}}$ 1
- (d) Use algebraic techniques to express $0.\dot{1}\dot{7}$ as a fraction in simplest terms. 3
- (e) The speed of light can be approximated as 3×10^8 metres per second. Calculate the time that it would take light to travel from the surface of a cricket ball onto the retina of a spectator sitting 80 metres away. Express your answer in scientific notation to 3 significant figures. 3
- (f) If the cost of an mp3 player is \$245.30 including 10% GST, then how much is the cost of the mp3 player without GST? 2

QUESTION TWO (23 Marks) **Start a new page**

(a) Expand and simplify the following:

(i) $(3x - 1)(2x + 3)$ 1

(ii) $(5 - x)^2$ 1

(b) Factorise:

(i) $y^2 - y - 90$ 1

(ii) $12x^2 + 16x - 3$ 2

(iii) $x^4 + 8x$ 3

(c) Simplify:

(i) $\frac{5}{4k} + \frac{k+1}{k^2}$ 2

(ii) $\frac{x^2 - 1}{x^3 - 1}$ 3

(d) Simplify fully:

(i) $\sqrt{5}(2\sqrt{3} - 5)$ 1

(ii) $5\sqrt{27} - 2\sqrt{75}$ 2

(e) Evaluate a and b if $(2\sqrt{3} + 1)^2 = a + \sqrt{b}$ given that a and b are integers. 3

(f) Express with a rational denominator (in simplest form):

(i) $\frac{5\sqrt{7}}{3\sqrt{2}}$ 1

(ii) $\frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ 3

QUESTION THREE (20 Marks) **Start a new page**(a) Solve for x

$$(i) \quad 4(3x - 2) - 5 = 35 \quad 2$$

$$(ii) \quad \frac{6}{x} = \frac{4}{5} \quad (x \neq 0) \quad 1$$

$$(iii) \quad 5x - 3 \geq -13 \quad 2$$

$$(iv) \quad \frac{x+4}{5} - \frac{x+2}{3} = -2 \quad 3$$

(b) The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$. Calculate the value of r when the volume of the sphere is 200cm^3 . Give your answer correct to 2 decimal places. 3

(c) Solve for x

$$(i) \quad x^2 - 5x = 0 \quad 2$$

$$(ii) \quad 4x^2 - 9 = 0 \quad 2$$

$$(iii) \quad 6x^2 + 13x + 5 = 0 \quad 3$$

(d) Solve for x correct to 3 decimal places:

$$x^2 - 5x + 2 = 0 \quad 2$$

END OF PAPER

Question 1

(a) 0.009700

✓ correct

✓ sig figs

(b)(i) $16b^{12}$

✓

(ii) $y^0 = 1$

✓

(c)(i) $\sqrt[3]{x^2}$ or $(\sqrt[3]{x})^2$

✓

(ii) $\frac{1}{\sqrt{5x-3}}$

✓

(d) let $x = 0.1717...$

$\therefore 100x = 17.1717...$

✓

$\therefore 99x = 17$

✓

$\therefore x = 17/99$

✓

(e) $t = \frac{80}{3 \times 10^8}$

✓

$= 0.000000026$

✓

$= 2.67 \times 10^{-7}$

✓

(f) 110% is \$245.30

10% is \$22.30

✓

100% is \$223

✓

Question 2

(a) (i) $6x^2 + 9x - 2x - 3$

$$= 6x^2 + 7x - 3$$

(ii) $25 - 10x + x^2$

(b) (i) $(y-10)(y+9)$

(ii) $12x^2 + 16x - 3$

$$= 12x^2 + 18x - 2x - 3$$

$$= 6x(2x+3) - (2x+3)$$

$$= (2x+3)(6x-1)$$

(iii) $x(x^3+8)$

$$= x(x+2)(x^2-2x+4)$$

(c) (i) $\frac{5k}{4k^2} + \frac{4k+4}{4k^2}$

$$= \frac{9k+4}{4k^2}$$

(ii) $\frac{(x-1)(x+1)}{(x-1)(x^2+x+1)}$

$$= \frac{x+1}{x^2+x+1}$$

(d) (i) $2\sqrt{5} - 5\sqrt{5}$

(ii) $15\sqrt{3} - 10\sqrt{3}$

$$= 5\sqrt{3}$$

(e) $(2\sqrt{3}+1)^2 = a + \sqrt{b}$

$$\therefore 12 + 4\sqrt{3} + 1 = a + \sqrt{b}$$

$$\therefore 13 + 4\sqrt{3} = a + \sqrt{b}$$

$$\therefore 13 + \sqrt{48} = a + \sqrt{b}$$

$$\therefore a = 13, b = 48$$

(f) (i) $\frac{5\sqrt{14}}{6}$

(ii) $\frac{(\sqrt{5}+1)(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)}$

$$= \frac{5+2\sqrt{5}+1}{5-1}$$

$$= \frac{6+2\sqrt{5}}{4}$$

$$= \frac{3+\sqrt{5}}{2}$$

Question 3

(a)(i) $12x - 8 - 5 = 35$ ✓

$$12x = 48$$

$$x = 12$$
 ✓

(ii) $30 = 4x$

$$x = 7.5$$
 ✓

(iii) $5x \geq -10$

$$x \geq -2$$
 ✓

(iv) $\frac{3x+12-(5x+10)}{15} = -2$ ✓

$$3x+12-5x-10 = -30$$

$$-2x = -32$$
 ✓

$$x = 16$$
 ✓

(b) $200 = \frac{4\pi r^3}{3}$ ✓

$$\frac{150}{\pi} = r^3$$

$$r = \sqrt[3]{\frac{150}{\pi}}$$
 ✓

$$= 3.63 \text{ (2 d.p.)}$$
 ✓

(c)(i) $x(x-5) = 0$ ✓

$$x = 0 \text{ or } 5$$
 ✓

(ii) $(2x-3)(2x+3) = 0$ ✓

$$x = \pm \frac{3}{2}$$
 ✓

(iii) $6x^2 + 10x + 3x + 5 = 0$ ✓

$$2x(3x+5) + (3x+5) = 0$$

$$(3x+5)(2x+1) = 0$$
 ✓

$$x = -\frac{5}{3} \text{ or } -\frac{1}{2}$$
 ✓

(d) $x = \frac{5 \pm \sqrt{25-8}}{2}$ ✓

$$= \frac{5 \pm \sqrt{17}}{2}$$

$$= 4.562 \text{ or } 0.438$$
 ✓